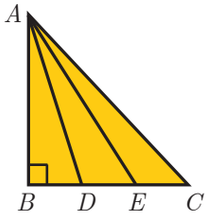
7. The perpendicular PS on the base QR of a ΔPQR intersects QR at S, such that $QS = 3\ SR$. Prove that $2PQ^2 = 2PR^2 + QR^2$

8. In the adjacent figure, ABC is a right angled triangle with right angle at B and points D, E trisect BC. Prove that $8AE^2 = 3AC^2 + 5AD^2$



4.5 Circles and Tangents

In our day-to-day real life situations, we have seen two lines intersect at a point or do not intersect in a plane. For example, two parallel lines in a railway track, do not intersect. Whereas, grills in a window intersect.

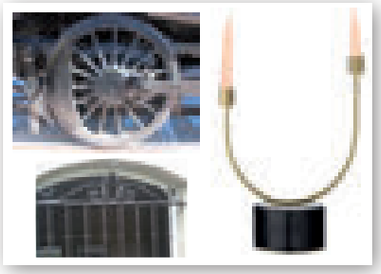
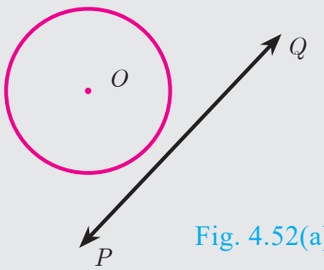
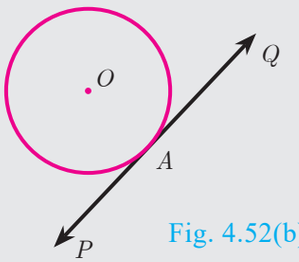
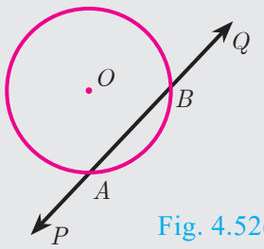


Fig. 4.51

Similarly what happens when a curve and a line is given in a plane? The curve may be parabola, circle or any general curve.

Similarly, what happens when we consider intersection of a line and a circle?

We may get three situations as given in the following diagram

Figure 1	Figure 2	Figure 3
 Fig. 4.52(a)	 Fig. 4.52(b)	 Fig. 4.52(c)
(i) Straight line PQ does not touch the circle.	(i) Straight line PQ touches the circle at a common point A.	(i) Straight line PQ intersects the circle at two points A and B.
(ii) There is no common point between the straight line and circle.	(ii) PQ is called the tangent to the circle at A.	(ii) The line PQ is called a secant of the circle.
(iii) Thus the number of points of intersection of a line and circle is zero.	(iii) Thus the number of points of intersection of a line and circle is one.	(iii) Thus the number of points of intersection of a line and circle is two.

Note

The line segment AB inscribed in the circle in Fig.4.52(c) is called chord of the circle. Thus a chord is a sub-section of a secant.

DO YOU KNOW?

The word "tangent" comes from the latin word "tangere" which means "to touch" and was introduced by Danish mathematician, 'Thomas Fineko' in 1583.

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